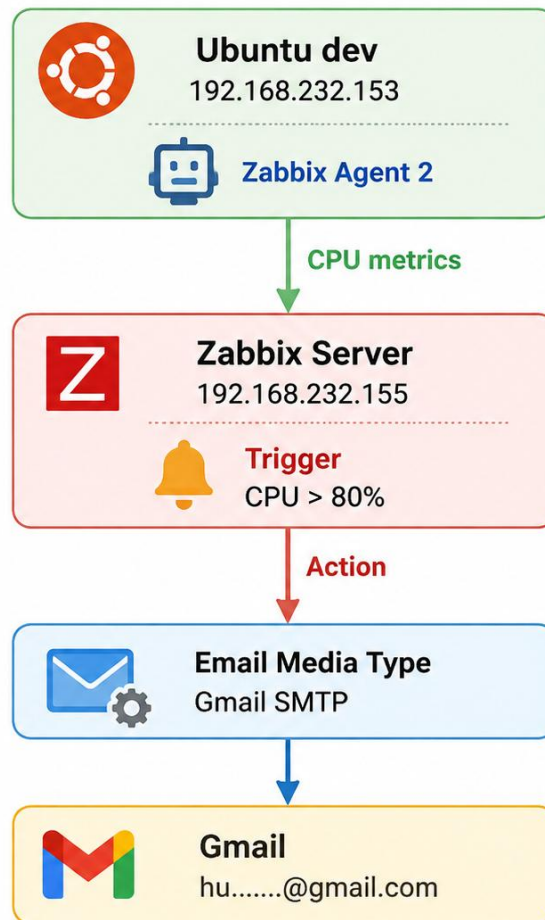


Sơ đồ cấu hình



Cấu hình zabbix alert qua gmail: Trigger cảnh báo khi cpu > 80%

Datacollection → hosts → trigger

The screenshot shows the Zabbix web interface with the **Hosts** configuration page. The left sidebar contains navigation links for Dashboards, Monitoring, Services, Inventory, Reports, and Data collection. The main content area displays the **Hosts** configuration form, including fields for Host groups, Templates, Name, DNS, IP, and Port. Below the form is a table listing the configured hosts.

Name	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption
dev	75	31	16	3	Web	192.168.232.153 10050		Linux by Zabbix agent	Enabled	ZBX	None
Zabbix server	153	83	16	6	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None

Tạo trigger cho dev:

Trigger

TriggerTagsDependencies

NameCPU usage is too high on app01

Event nameCPU usage is too high on app01

Operational data

SeverityNot classifiedInformationWarningAverageHighDisaster

Expressionlast(/dev/system.cpu.util)>80Add

Expression constructor

OK event generationExpressionRecovery expressionNone

PROBLEM event generation modeSingleMultiple

OK event closesAll problemsAll problems if tag values match

Allow manual close

Menu entry nameTrigger URL

UpdateCloneDeleteCancel

Severity	Value	Name	Operational data	Expression	Status	Info	Tags
High	OK	CPU usage is too high on app01		last(/dev/system.cpu.util)>80	Enabled		

Làm quá tải (stress) máy app1 để tải cao trong 2 phút

```
cluster4@app01:~$ stress --cpu 2 --timeout 120
stress: info: [1956] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
stress: info: [1956] successful run completed in 120s
cluster4@app01:~$
```

Kiểm tra trigger trên zabbix

Chờ khoảng 30s nếu cpu quá 80% hệ thống sẽ cảnh báo

Vào monitoring → problems | status = PROBLEM

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
06:44:28 AM	High		PROBLEM		dev	CPU usage is too high on app01	44s	Update		class:os component:cpu target:linux

Displaying 1 of 1 found

Sau khi stress kết thúc, cpu < 80% zabbix chuyển status = RESOLVED

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
06:44:28 AM	High	06:45:28 AM	RESOLVED		dev	CPU usage is too high on app01	1m	Update		class:os component:cpu target:linux

Displaying 1 of 1 found

Cấu hình trigger và gửi mail cảnh báo trên

Chuẩn bị 2 email mail gửi/mail nhận

Vào mục alert → media types → Email

Cấu hình mail gửi đi

Media type

Media type Message templates 5 Options

* Name	Email
Type	Email
Email provider	Generic SMTP
* SMTP server	smtp.gmail.com
SMTP server port	587
* Email	hungdoansky@gmail.com
SMTP helo	gmail.com
Connection security	None STARTTLS SSL/TLS
SSL verify peer	☐
SSL verify host	☐
Authentication	None Username and password
Username	hungdoansky@gmail.com
Password	

Update Clone Delete Cancel

Test mail cho mail nhận

Test media type "Email" ✕

✓

Media type test successful. ✕

* Send to

hungdoan370@gmail.com

Subject

Test subject

* Message

This is the test message from Zabbix

Test

Cancel

cấu hình Email cho User nhận cảnh báo

vào User → user → admin → media: cấu hình thông tin mail nhận

Media ✕

Type

Email ▼

* Send to

hungdoan370@gmail.com

[Remove](#)

[Add](#)

* When active

1-7,00:00-24:00

Use if severity

☐ Not classified

☒ Information

☒ Warning

☒ Average

☒ High

☒ Disaster

Enabled

☒

Add

Cancel

Media	Type	Send to	When active	Use if severity	Status	Action
	Email	hungdoan370@gmail.com	1-7,00:00-24:00	N I W A H D	Enabled	Edit Remove
Add						

[Update](#)
[Delete](#)
[Cancel](#)

Add trigger

Vào Alerts → Actions → Trigger actions: tạo trigger nhấn add

New condition

Type

Trigger

Operator

equals

does not equal

Trigger source

Host

Template

* Triggers

dev: CPU usage is too high on app01

×

type here to search

Select

Add

Cancel

Vào tab operation nhấn add:

Cấu hình đúng user admin

☐	Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update
☐	07:45:28 AM	High		PROBLEM		dev	CPU usage is too high on app01	11s	Update
☐	07:06:40 AM	Warning		PROBLEM		dev	Linux: Number of installed packages has been changed	38m 59s	Update
☐	07:06:39 AM	Information		PROBLEM		dev	Linux: Operating system description has changed	39m	Update

Kiểm tra mail nhận → trigger quá tải = gửi mail

Problem: CPU usage is too high on app01 Hộp thư đến x



hungdoansky@gmail.com

đến tôi ▾

Problem started at 07:45:28 on 2026.08.07

Problem name: CPU usage is too high on app01

Host: dev

Severity: High

Operational data: 99.99 %

Original problem ID: 108

← Trả lời

→ Chuyển tiếp



Sau khi tắt quá trình tải cpu

```
cluster4@app01:~$ stress-ng --cpu 2 --timeout 5m
stress-ng: info: [3015] setting to a 5 mins, 0 secs run per stressor
stress-ng: info: [3015] dispatching hogs: 2 cpu
^Cstress-ng: info: [3015] skipped: 0
stress-ng: info: [3015] passed: 2: cpu (2)
stress-ng: info: [3015] failed: 0
stress-ng: info: [3015] metrics untrustworthy: 0
stress-ng: info: [3015] successful run completed in 2 mins, 34.76 secs
```

Trigger báo status = resolved

☐	Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
☐	07:45:28 AM	High	07:47:28 AM	RESOLVED		dev	CPU usage is too high on app01	2m	Update		class:os component:cpu target:linux
☐	07:06:40 AM	Warning		PROBLEM		dev	Linux: Number of installed packages has been changed	40m 48s	Update		class:os component:os scope:notice **
☐	07:06:39 AM	Information		PROBLEM		dev	Linux: Operating system description has changed	40m 49s	Update		class:os component:os scope:notice **

Cấu hình mail nhận recovery operation/ sau khi máy đã khôi phục

Vào arlert → trigger action

Cấu hình vừa rồi

Trigger actions ? [Create action](#)

Status Any Enabled Disabled Filter

[Apply](#) [Reset](#)

☐ Name ▲	Conditions	Operations	Status	Info
☐ Name: CPUAlert - Send Email	Trigger equals dev: CPU usage is too high on app01	Send message to users: Admin (Zabbix Administrator) via Email	Enabled	

Vào Recovery operation nhấn add:

Action ? ×

[Action](#) [Operations 2](#)

* Default operation step duration

Operations	Steps	Details	Start in	Duration	Action
	1	Send message to users: Admin (Zabbix Administrator) via Email	Immediately	Default	Edit Remove
	Add				

Recovery operations	Details	Action
	Send message to users: Admin (Zabbix Administrator) via Email	Edit Remove
	Add	

Update operations	Details	Action
	Add	

☒ Pause operations for symptom problems

☒ Pause operations for suppressed problems

☒ Notify about canceled escalations

* At least one operation must exist.

[Update](#) [Clone](#) [Delete](#) [Cancel](#)

Cấu hình mail nhận khi recovery

Operation details

×

Operation

Send message

▼

*

At least one user or user group must be selected.

Send to user groups

type here to search

Select

Send to users

Admin (Zabbix Administrator) ×

type here to search

Select

Send to media type

Email

▼

Custom message

☒

Subject

[RECOVERY] CPU đã trở lại bình thường - app01

Message

ZABBIX RECOVERY
Host: {HOST.NAME}
IP: {HOST.IP}
Problem: {EVENT.NAME}

Update

Cancel

Lưu cấu hình

Trigger actions

?

Create action

✓

Action updated

×

Filter

Name

Status

Any

Enabled

Disabled

Apply

Reset

Name	Conditions	Operations	Status	Info
☐ Name: CPUAlert - Send Email	Trigger equals dev: CPU usage is too high on app01	Send message to users: Admin (Zabbix Administrator) via Email	Enabled	

Test quá tải cpu

Vào máy app01 chạy lệnh làm quá tải cpu

```
cluster4@app01:~$ stress-ng --cpu 2 --timeout 5m
stress-ng: info:  [3015] setting to a 5 mins, 0 secs run per stressor
stress-ng: info:  [3015] dispatching hogs: 2 cpu
```

☐	Time ▼	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
☐	07:54:28 AM	High		PROBLEM		dev	CPU usage is too high on app01	1s	Update		class: os component: cpu target: linux
☐	07:06:40 AM	Warning		PROBLEM		dev	Linux: Number of installed packages has been changed	47m 49s	Update		class: os component: os scope: notice ***
☐	07:06:39 AM	Information		PROBLEM		dev	Linux: Operating system description has changed	47m 50s	Update		class: os component: os scope: notice ***

Displaying 3 of 31

Problem: CPU usage is too high on app01

Hộp thư đến x



hungdoansky@gmail.com

đến tôi ▼



Có vẻ như thư này được viết bằng Tiếng Anh

[Dịch sang Tiếng Việt](#)

Problem started at 07:54:28 on 2026.08.07

Problem name: CPU usage is too high on app01

Host: dev

Severity: High

Operational data: 99.1 %

Original problem ID: 110

Tắt quá trình overload cpu

☐	Time ▼	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
☐	07:54:28 AM	High	07:55:28 AM	RESOLVED		dev	CPU usage is too high on app01	1m	Update		class: os component: cpu target: linux
☐	07:06:40 AM	Warning		PROBLEM		dev	Linux: Number of installed packages has been changed	48m 48s	Update		class: os component: os scope: notice ***
☐	07:06:39 AM	Information		PROBLEM		dev	Linux: Operating system description has changed	48m 49s	Update		class: os component: os scope: notice ***

[RECOVERY] CPU đã trở lại bình thường - app01 Hộp thư đến x



hungdoansky@gmail.com
đến tôi ▾

14:55 {

 Có vẻ như thư này được viết bằng Tiếng Anh
[Dịch sang Tiếng Việt](#) ✕

ZABBIX RECOVERY

Host: dev
IP: 192.168.232.153

Problem: CPU usage is too high on app01

CPU đã trở lại bình thường.

Current value: 54.54 %

Test thành công gửi và nhận mail cảnh báo khi cpu > 80%

